

International Finance

Lecture IX – April 14th

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. International transfers/payments. Channels, related instruments. (February 24th)
3. Balance of payments structure and national income accounting for an open economy. International investment position (March 3rd)
4. Introduction to exchange rate determination: asset approach. Covered and uncovered interest rate parity. (March 10th)
5. Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity. Overshooting. (March 17th)
6. Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect. Application on the nominal and real convergence in the EU (March 24th)
7. Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies. (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. **Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications.** Interventions and sterilisation. **Policy trilemma.** (April 14th)
10. **Monetary integration: costs and benefits. Monetary integration: effects – empirical research.** (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. **Reserved for a make-up class (possibly for April 21st). (May 19th)**

Outline

- Course organization
 - Attendance, quiz, assignments (Teams + topics)
 - Quiz #8: will be open by Thursday midnight
 - The extended deadline: Monday April 27th midnight
- Catching up: ML condition
 - What do we know about the elasticities
- Macro view: external and internal balance
 - “Impossible Trinity”
 - Swan diagram and Tinbergen’s rule

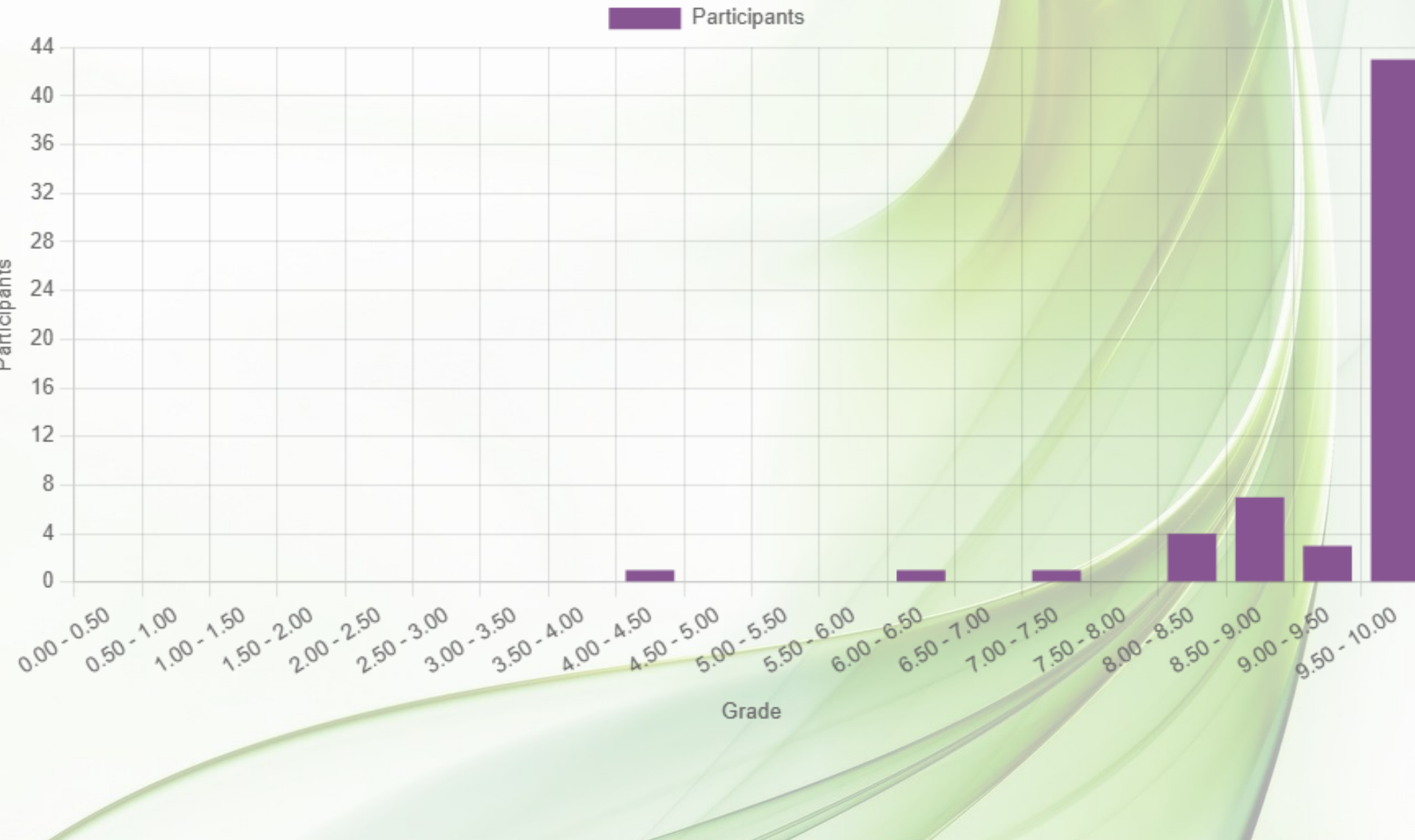
Next Week: April 21st

- No class on April 21st
- Instead, reading and slides will be released on Moodle
 - Topic: Monetary integration
 - A reading assignment (with questions)
 - It will be released by Friday
 - Questions will be discussed at the next session (April 28th)
 - Possibly in the form of a simple quick quiz

Quiz #8

Average: 9.48

64 attempts/63 participants



Reading for This Week:

- KOM – chapter 18: Fixed Exchange Rates and Foreign Exchange Intervention
- Frankel (2025): Resuscitating the Salter-Swan model of a small open economy
 - <https://cepr.org/voxeu/columns/resuscitating-salter-swan-model-small-open-economy>

Reading for the Next Two Weeks

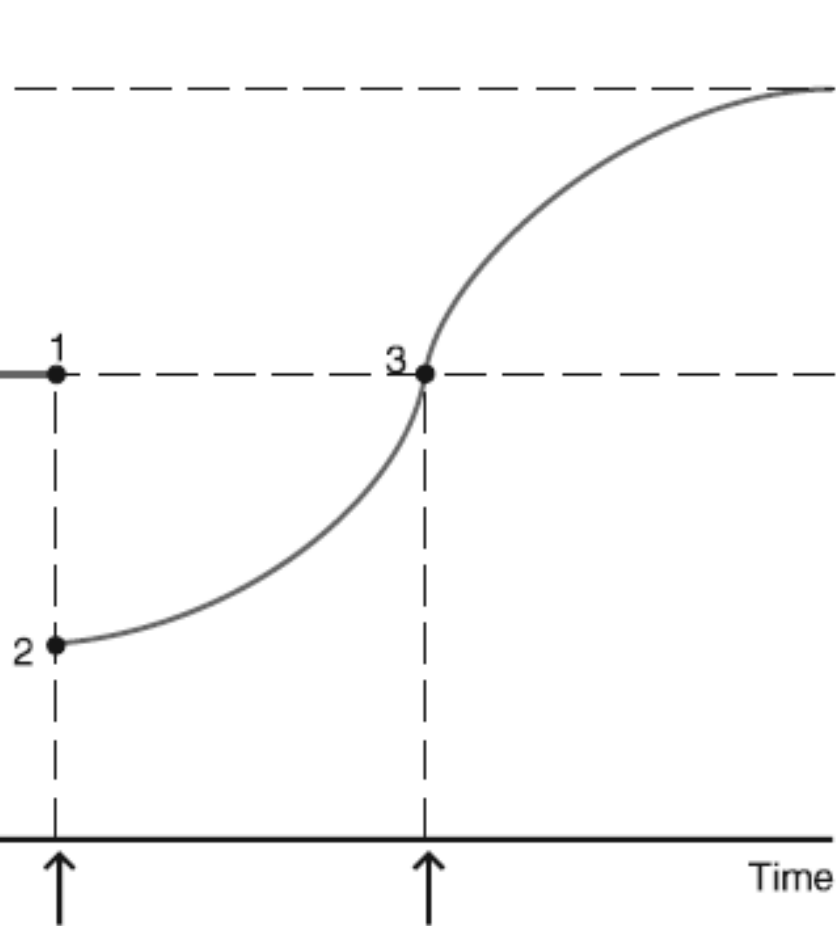
- Basic overview:
 - KOM - chapter 20: Optimum Currency Areas and the European Experience
 - For those more ambitious:
 - Beetsma & Giuliodori (2010): The Macroeconomic Costs and Benefits of the EMU and Other Monetary Unions: An Overview of Recent Research
- Texts on the OCA and EMU:
 - Mundell (1961): A Theory of Optimum Currency Areas
 - De Grauwe (1991): Costs and Benefits of a Monetary Union
- Texts on the effects of Euro introduction:
 - Brachinger (2006): Euro or “Teuro”? The Euro-induced Perceived Inflation in Germany
 - Falagardia et al (2023): Has the euro changeover really caused extra inflation in Croatia?
- Texts on the preparedness of Czechia:
 - CNB: Analyses of the Czech Republic’s Current Economic Alignment with the Euro Area – 2025

Marshall-Lerner Condition & J-Curve

J-Curve Effect

Current account (in domestic output units)

Long-run effect of real depreciation on the current account



Real depreciation takes place and J-curve begins

End of J-curve

Cause: time lags in producer and consumer responses

Devaluations and Trade Balance?

Marshall-Lerner Condition

- Elasticity approach to balance of payments
- Alfred Marshall, Abba Lerner, J. Robinson, F. Machlup
- Central question:
 - How does devaluation affect current account? Does the devaluation of a currency improve the country's balance of trade?
- **M-L condition:**
 - Condition that guarantees that devaluation will improve current account
- The condition is based on **elasticities**
 - Price elasticity shows the sensitivity of an economic variable to a change in price
 - E.g. price elasticity of demand for exports:

$$\eta_x = \frac{\frac{dX}{X}}{\frac{dS}{S}}$$

ML Condition - Mathematics

Normalize P and P* to 1

$$CA = P \cdot X_v - S \cdot P^* \cdot M_v$$

$$CA = X - S \cdot M$$

$$dCA = dX - dS \cdot M - S \cdot dM$$

$$\frac{dCA}{dS} = \frac{dX}{dS} - \frac{dS}{dS} \cdot M - S \cdot \frac{dM}{dS}$$

$$\frac{dCA}{dS} = \frac{\eta_x \cdot X}{S} + \eta_m \cdot M - M$$

$$\frac{dCA}{dS} \cdot \frac{1}{M} = \frac{\eta_x \cdot X}{S \cdot M} + \eta_m - 1$$



Elasticities:

$$\eta_x = \frac{\frac{dX}{dS}}{\frac{X}{S}} \quad \eta_m = -\frac{\frac{dM}{dS}}{\frac{M}{S}}$$

Additional assumption:

Balanced trade $\frac{X}{S \cdot M} = 1$

$$\frac{dCA}{dS} = M \cdot (\eta_x + \eta_m - 1)$$

ML Condition - Interpretation

- Devaluation will improve trade balance only if:

$$\eta_x + \eta_m > 1$$

- This means that export and import must be sufficiently elastic with respect to price!

Empirical Evidence on Elasticities

Country	Elasticity of export demand	Elasticity of import demand	Sum
USA	1.19	1.24	2.43
UK	0.86	0.65	1.51
France	1.28	0.93	2.21
Germany	1.02	0.79	1.81
Industrial countries average	1.11	0.99	2.10
India	0.5	2.2	2.7
Korea	2.5	0.8	3.3
Developing countries average	1.1	1.5	2.6

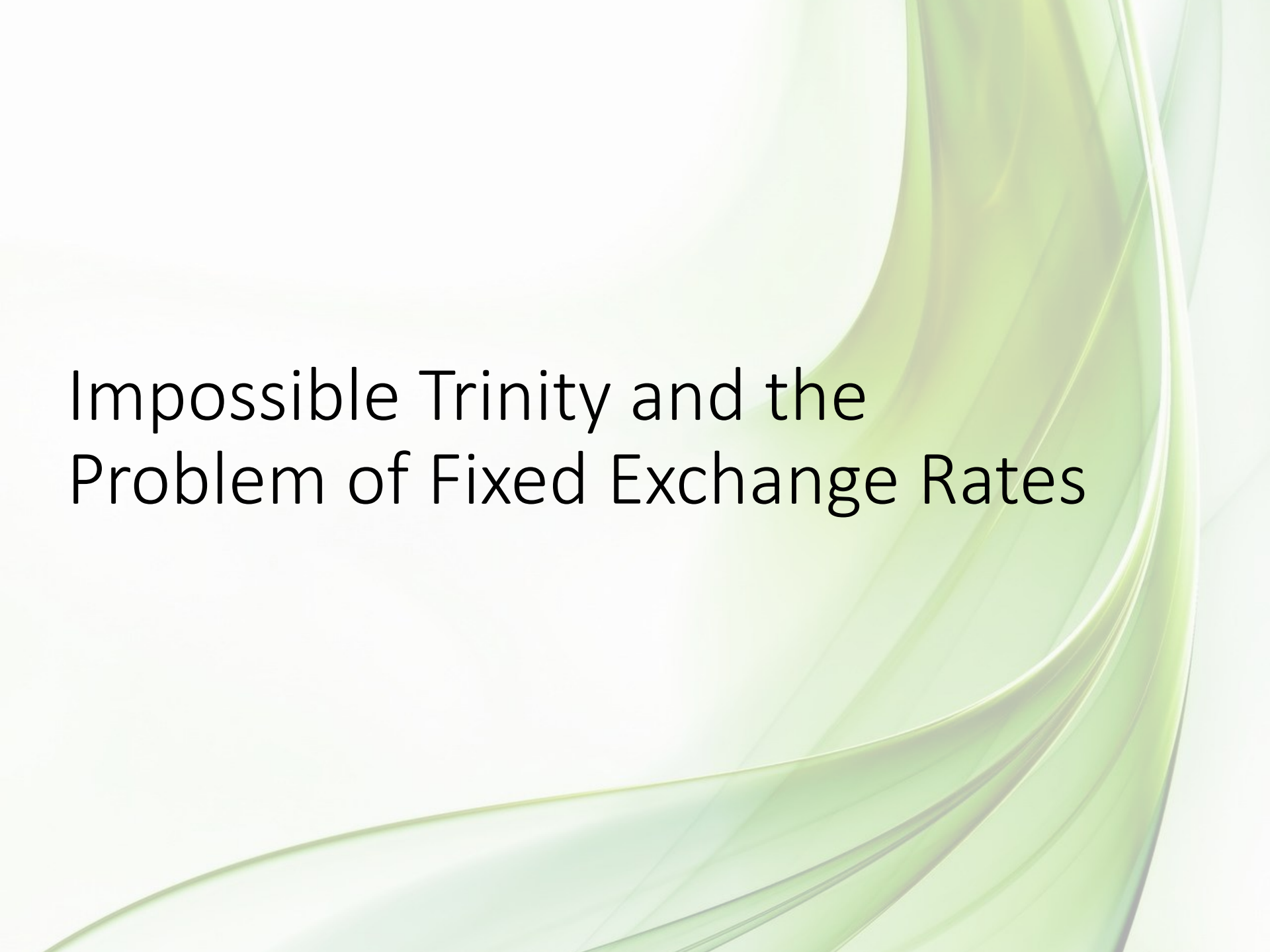
Elasticities on 2-3 year period, 1969-1981,
data from Pilbeam (1992), resp. Gylfasson (1987)

Additional Evidence?

- Rose (1991)
 - A variety of parametric and non-parametric techniques used on data for five OECD countries
 - *“There is little evidence that the exchange rate significantly affects the trade balance”.*
- Giudici & Lima (2026) – Brazilian data (but deeper):
 - We find significant evidence supporting the Marshall-Lerner condition across all sectors analysed.
 - Although we have not observed significant evidence supporting the J-curve, there is a higher probability of the phenomenon occurring than not occurring.
- Meta-analysis: Bahmani et al. (2013)
 - Literature review + own estimation of 29 countries' trade elasticities
 - Although the point estimates in many studies suggest that the M-L condition is met, it really is not met in half of the cases.
 - **Practical implications:** Policymakers who hope to improve their countries' competitive position could benefit from learning that this policy is indeed less effective than might be supposed.

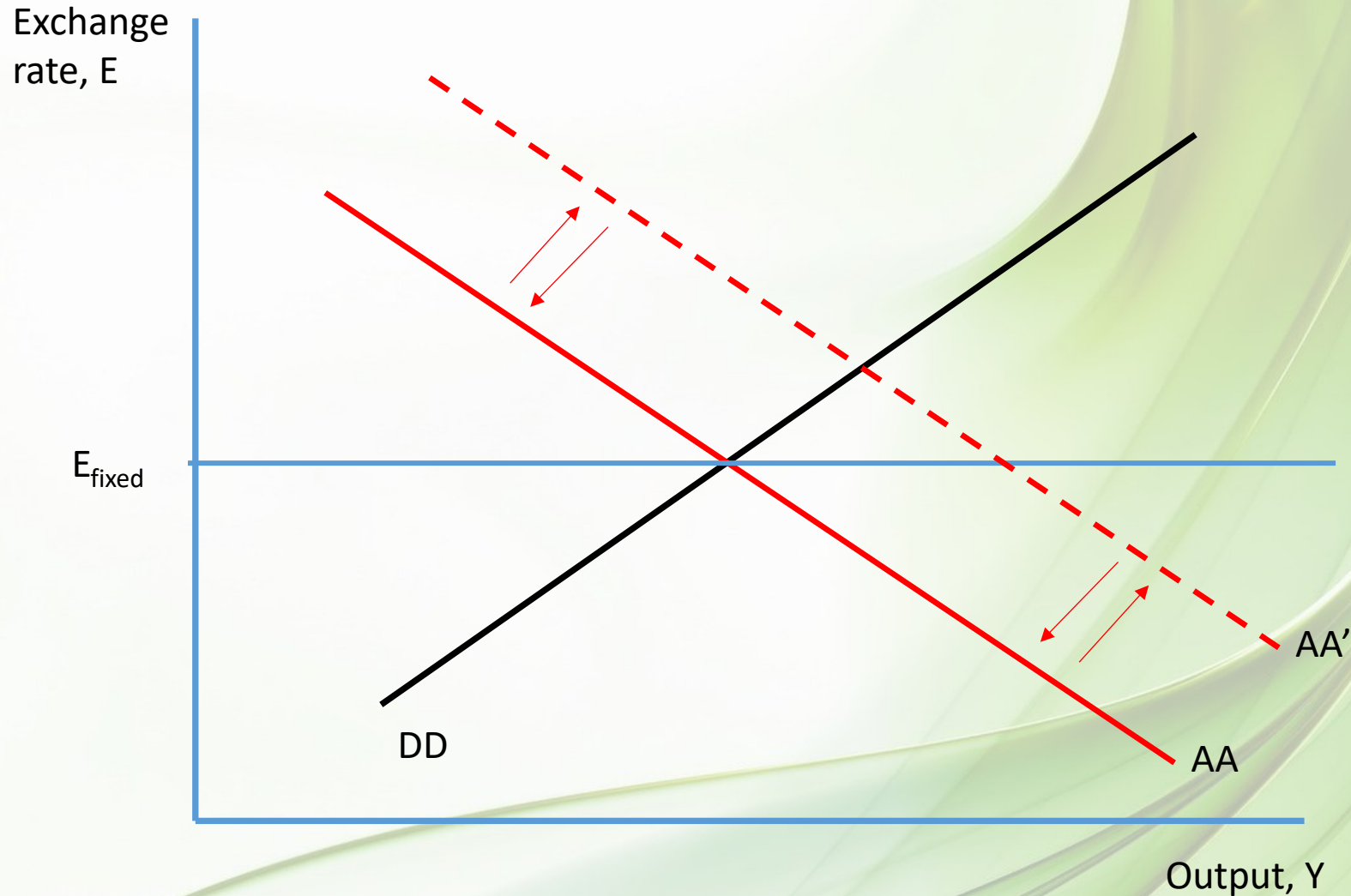
Summary

- If exports and imports adjust gradually to real exchange rate changes, the current account may follow a J-curve pattern after a real currency depreciation, first worsening and then improving.
- Evidence is mixed, but we often assume that long-run elasticities meet the ML condition.

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Impossible Trinity and the Problem of Fixed Exchange Rates

Fixed Exchange Rate: Monetary Expansion

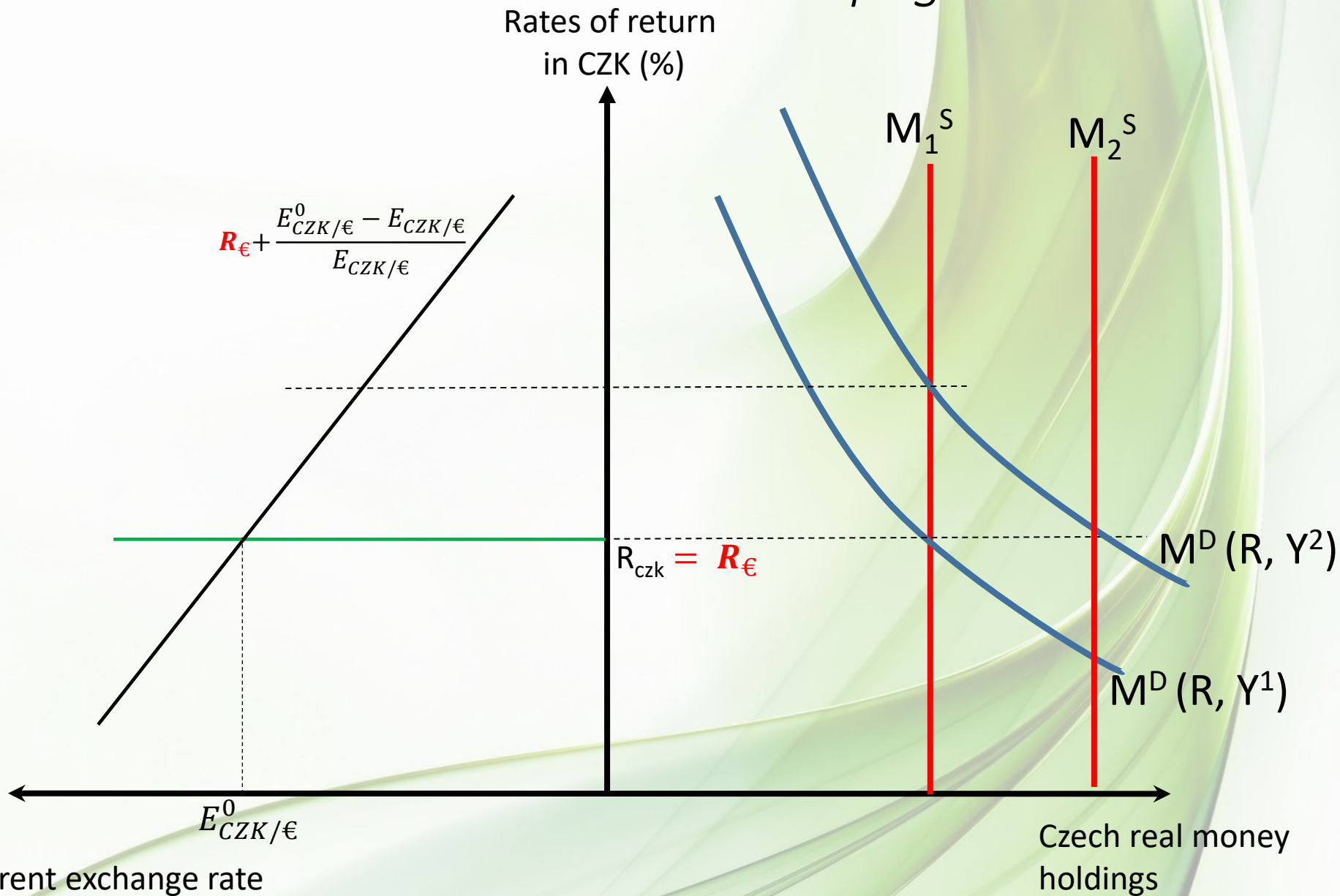


Monetary Expansion: Explanation

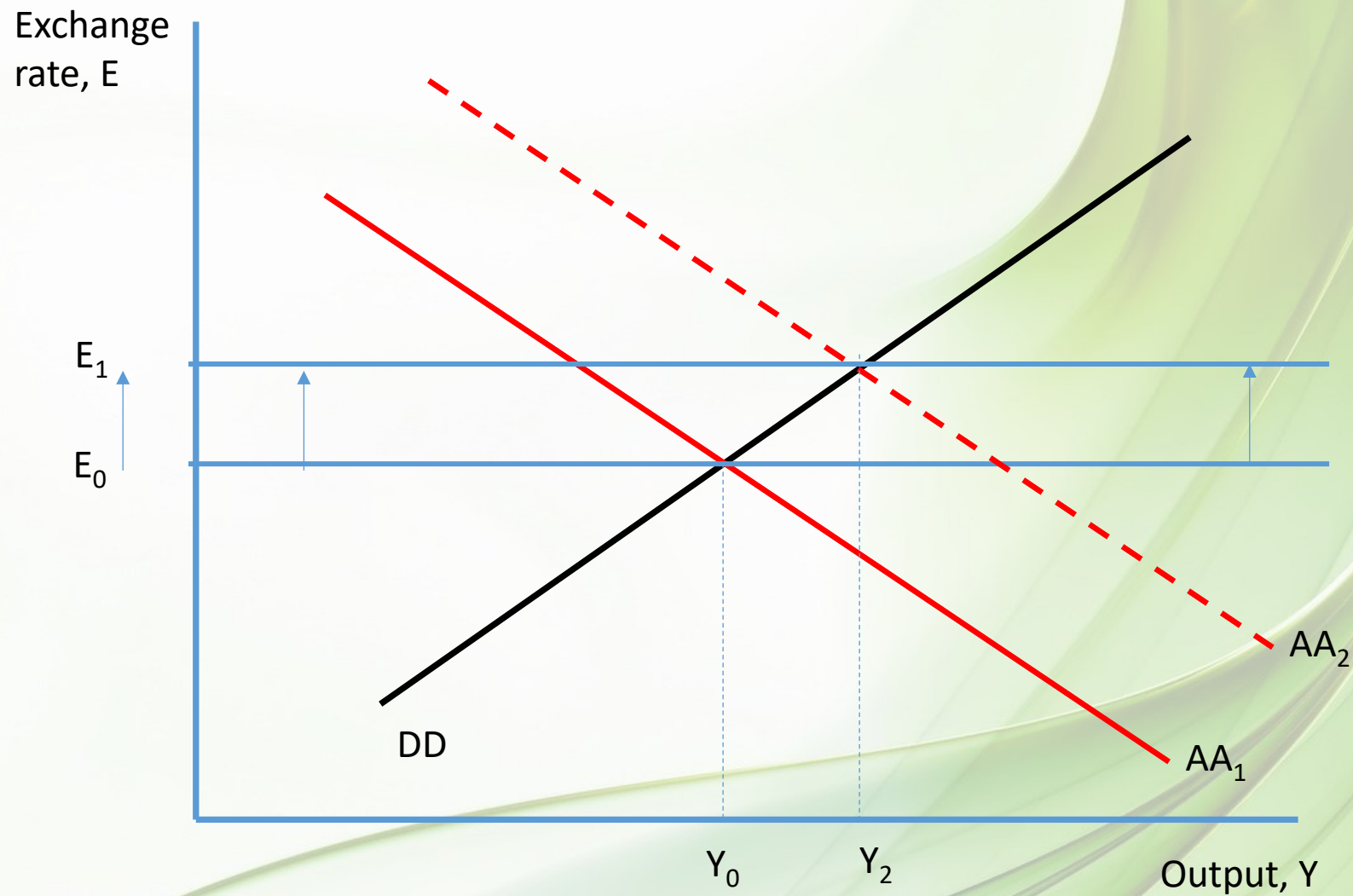
- Central bank attempts to increase money supply, the AA curve should move to the right
- This will reduce interest rates and it will lead to a pressure towards the depreciation of the domestic currency
- Central bank will defend the original exchange rate by intervening (selling foreign currency and buying the domestic currency)
- An unsterilized intervention will reduce the central bank's FOREX reserves and the monetary base
- The AA curve will be returning back to the original position
- **Monetary policy becomes impotent....**

Asset Market Equilibrium and Fixed ER

Assume that CZK would have the peg to Euro



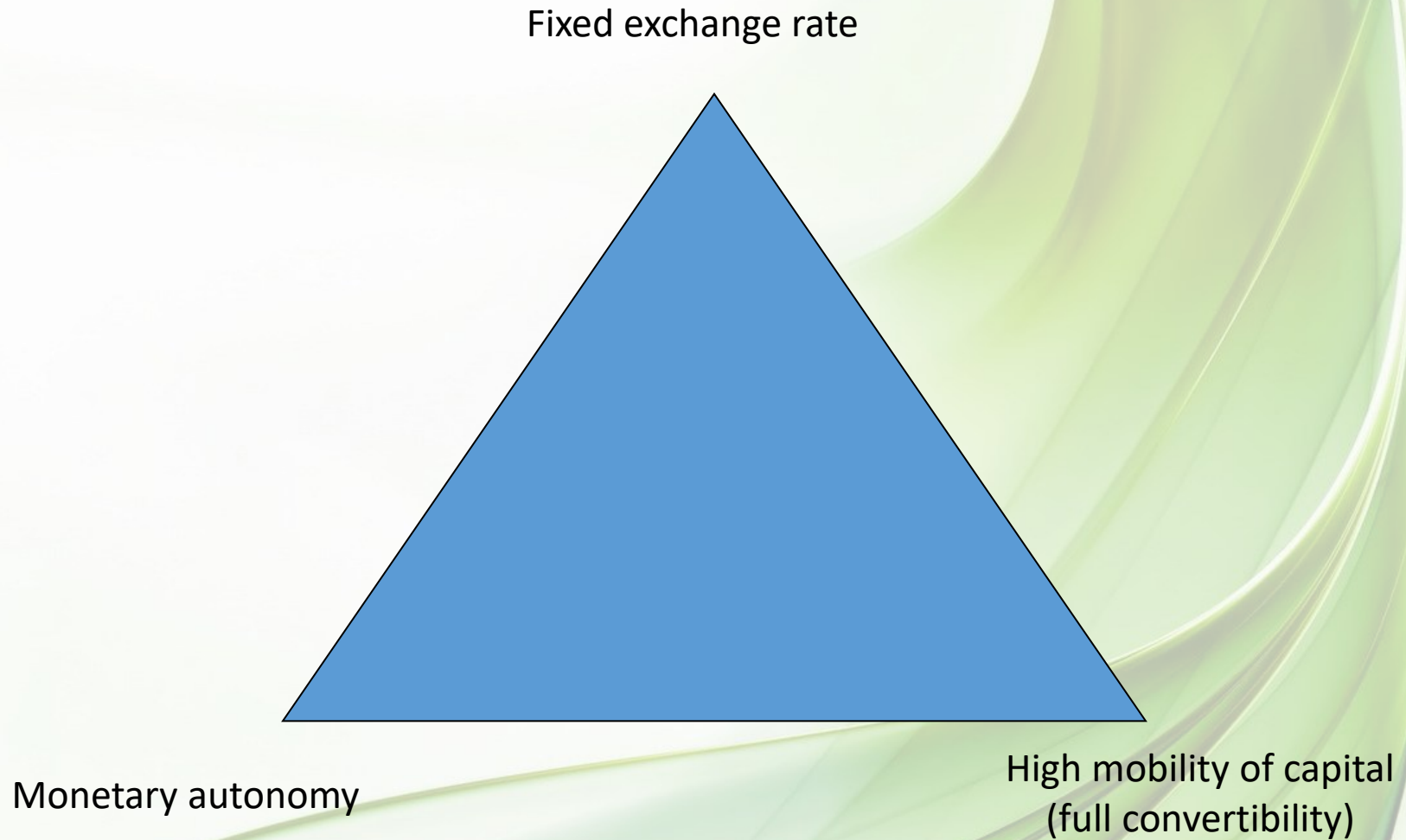
And How About a Devaluation?



Devaluation

- Exchange rate changes from E_0 to E_1
- Domestic goods and services become relatively cheaper (relative to foreign goods)
- Output increases ($Y \uparrow$)
- The new level of output not on the AA_1 curve
 - Excess money demand would be pushing up interest rates $\rightarrow$ central bank must intervene
 - Form of intervention: CEB buys foreign assets, money supply increases

Impossible Trinity and the Problem of Fixed Exchange Rates



You can just have two of these in the long run, never all three!

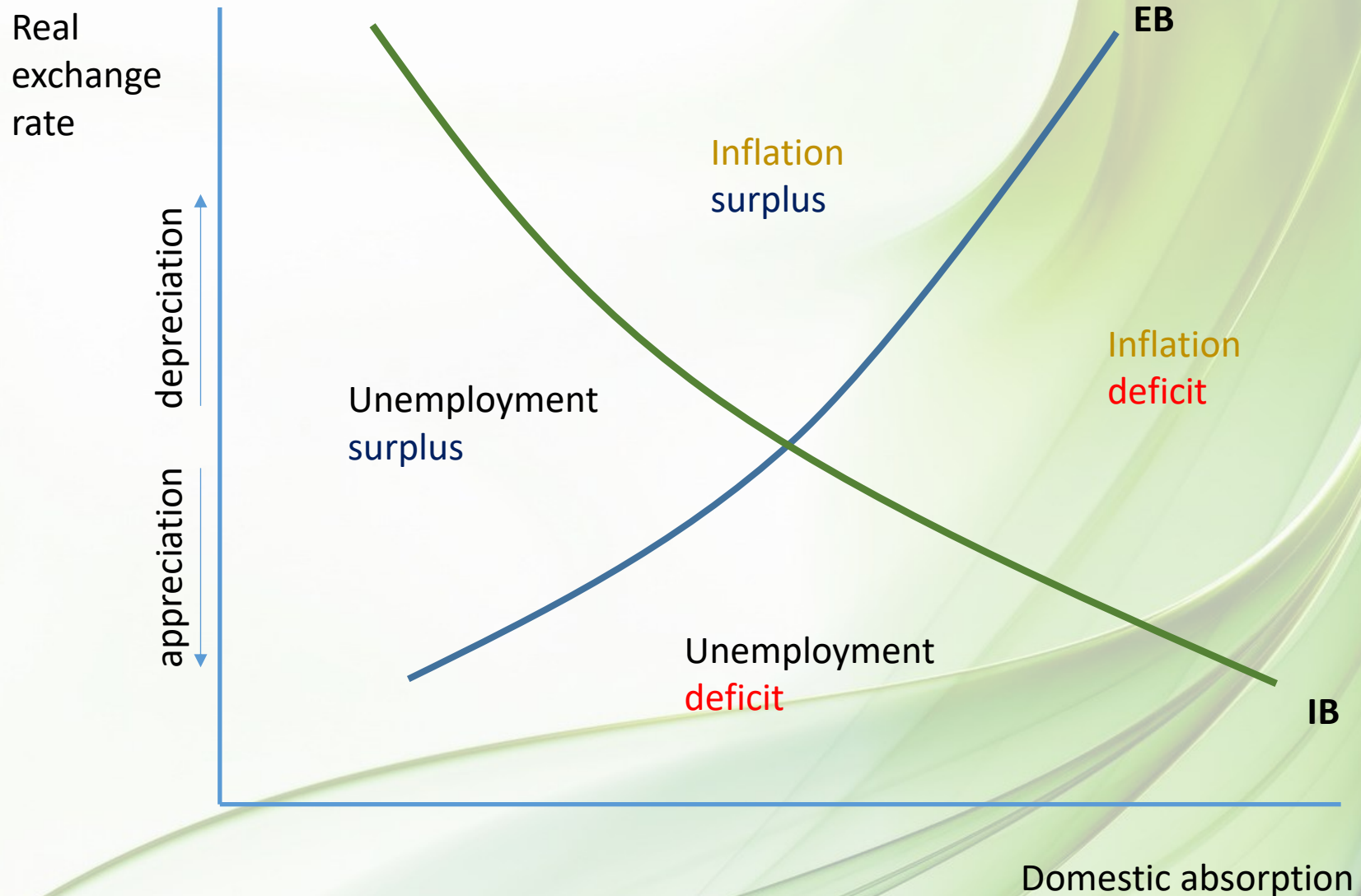
Swan Diagram & Tinbergen's Rules

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Economic Policy Approach

- Tinbergen (1952), Meade (1951)
- Devaluation as one of the two independent policy instruments required for achieving two policy objectives (internal and external equilibrium)
- Simplified (incorrect) interpretation:
 - Effects of devaluations depend on elasticities of demand for imports
- The actual focus:
 - The proper combination of policies to employ to achieve certain policy results
 - If there is mass unemployment, then other instrument is available, devaluation is not necessarily the only or the sound option.

Swan Diagram



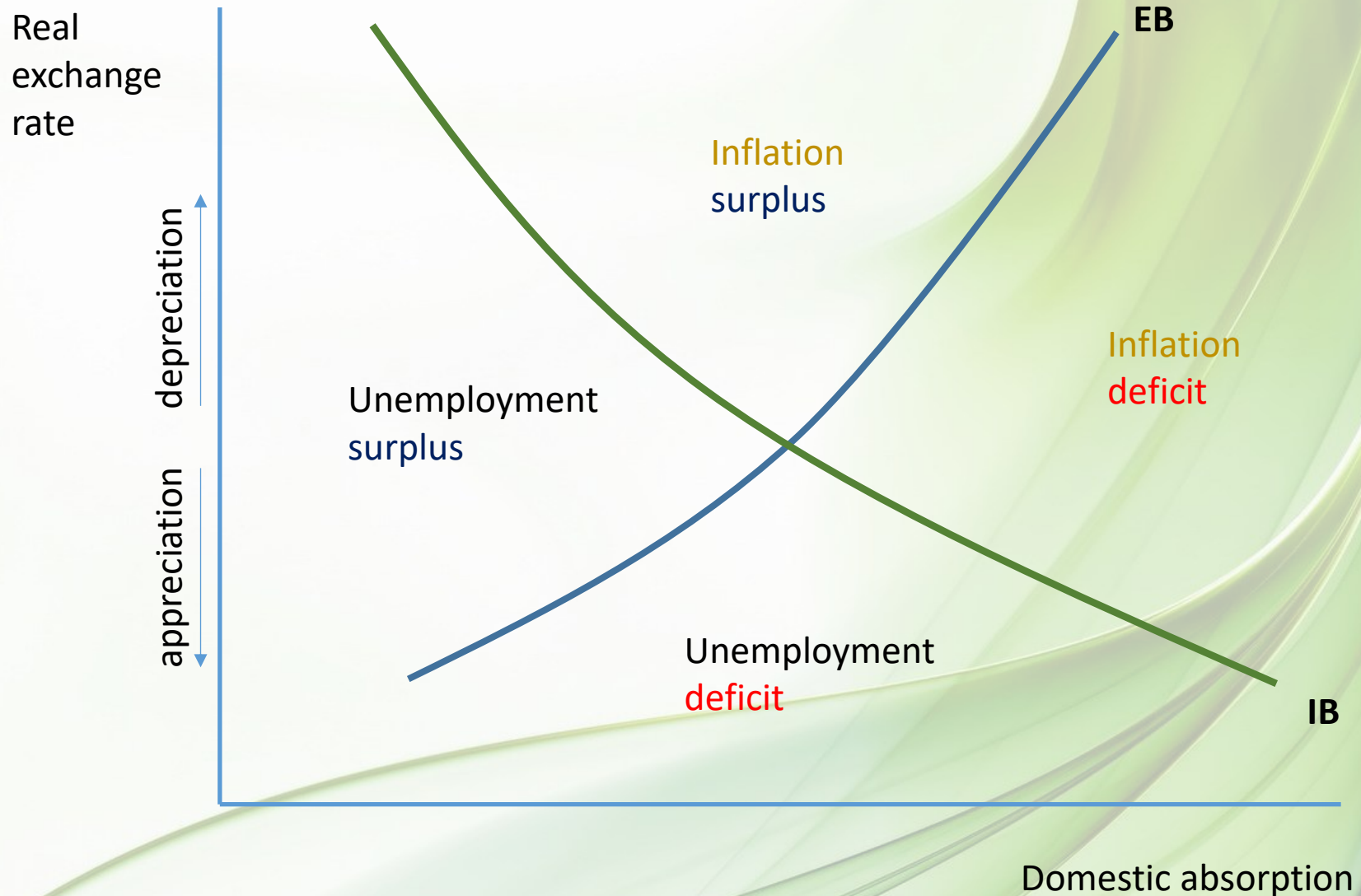
Internal and External Balance: Overview

- Historical perspective
 - Older literature (esp. 1948-1973) more focused on the relative efficiency of fiscal/monetary policies, devaluations/revaluations were viewed as a bit extreme measures under the Bretton Woods system
 - For cases of “fundamental disequilibrium”
- Policy objectives
 - Full employment + stable prices: **internal equilibrium**
 - Equilibrium in balance of payments (supply and demand for the currency): **external equilibrium**
- Policy classification:
 - **Expenditure changing** – influence the level of aggregate demand
 - **Expenditure switching** – influence the composition of spending
 - Examples: revaluation/devaluation
- **How can the authorities simultaneously achieve both internal and external equilibrium?**

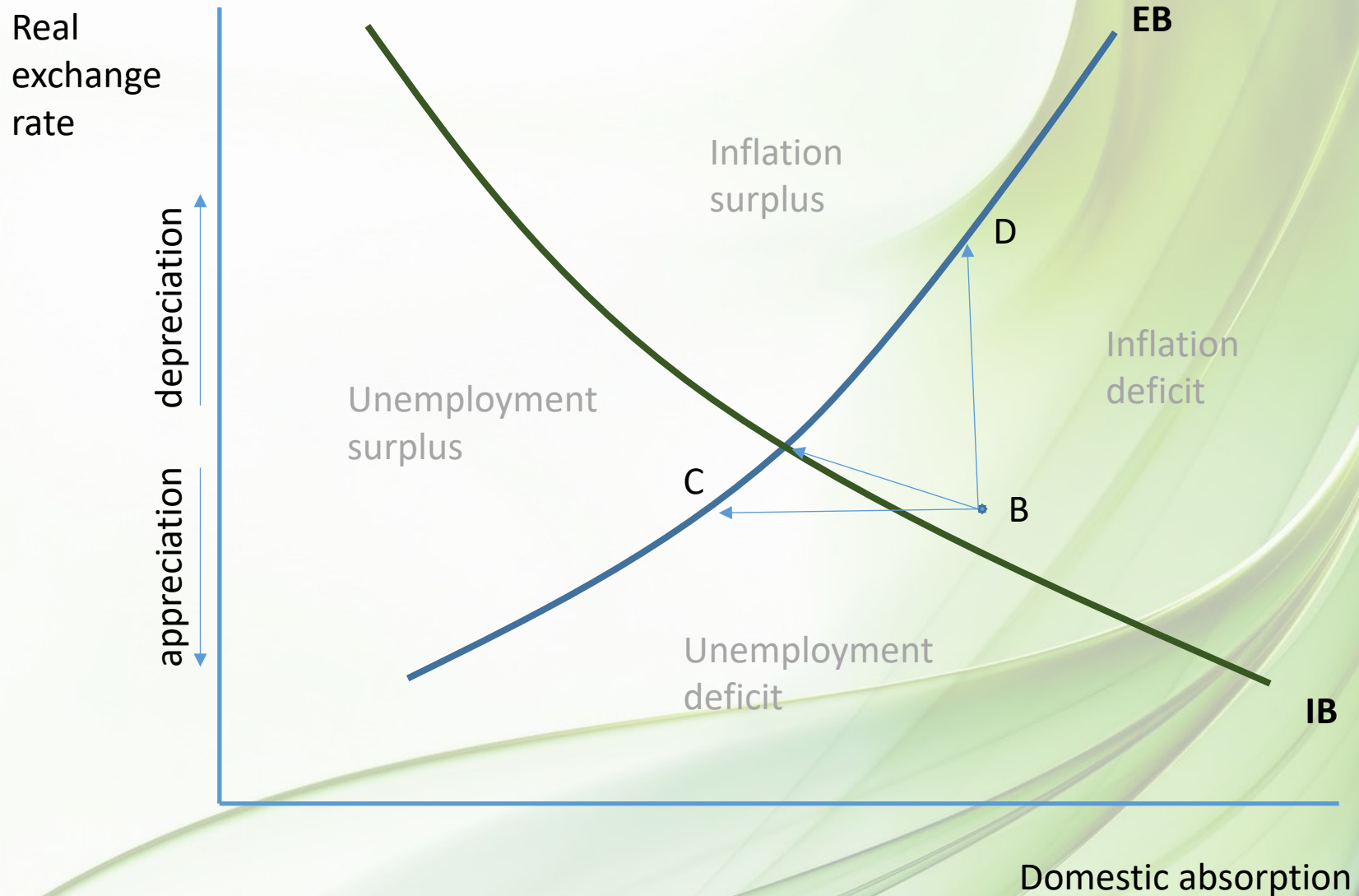
Swan Diagram

- Trevor Swan (1955)
- Axis:
 - Real exchange rate (rise means real depreciation)
 - Domestic absorption ($A = C + I + G$)
- **IB schedule**
 - Combination of the real exchange rate and domestic absorption for which the economy is in internal balance (full employment, stable prices)
 - Downward sloping
- **EB schedule**
 - Combination of the real exchange rate and domestic absorption for which the economy is in external balance (current account equilibrium)
 - Upward sloping

Swan Diagram



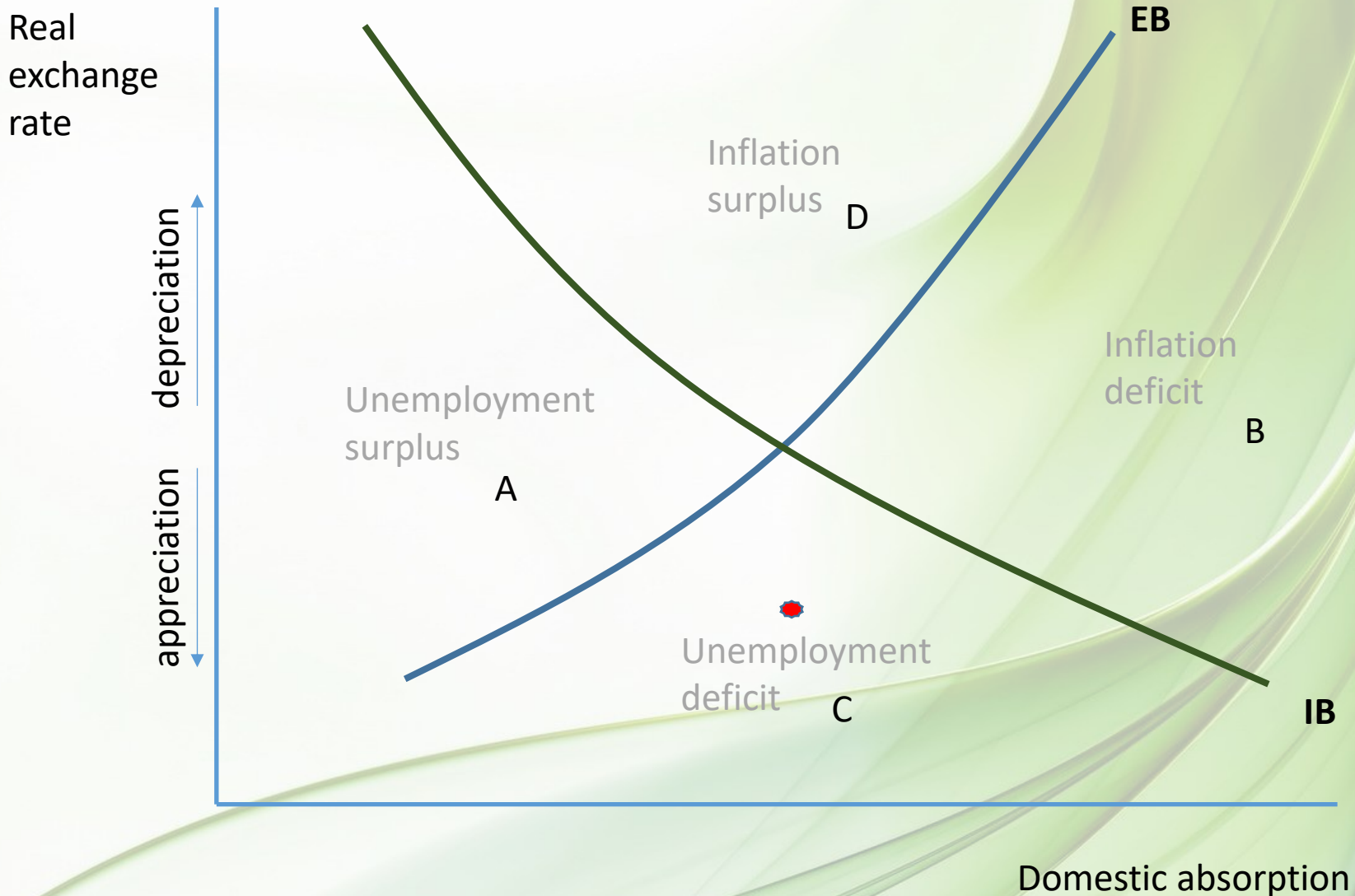
Swan Diagram

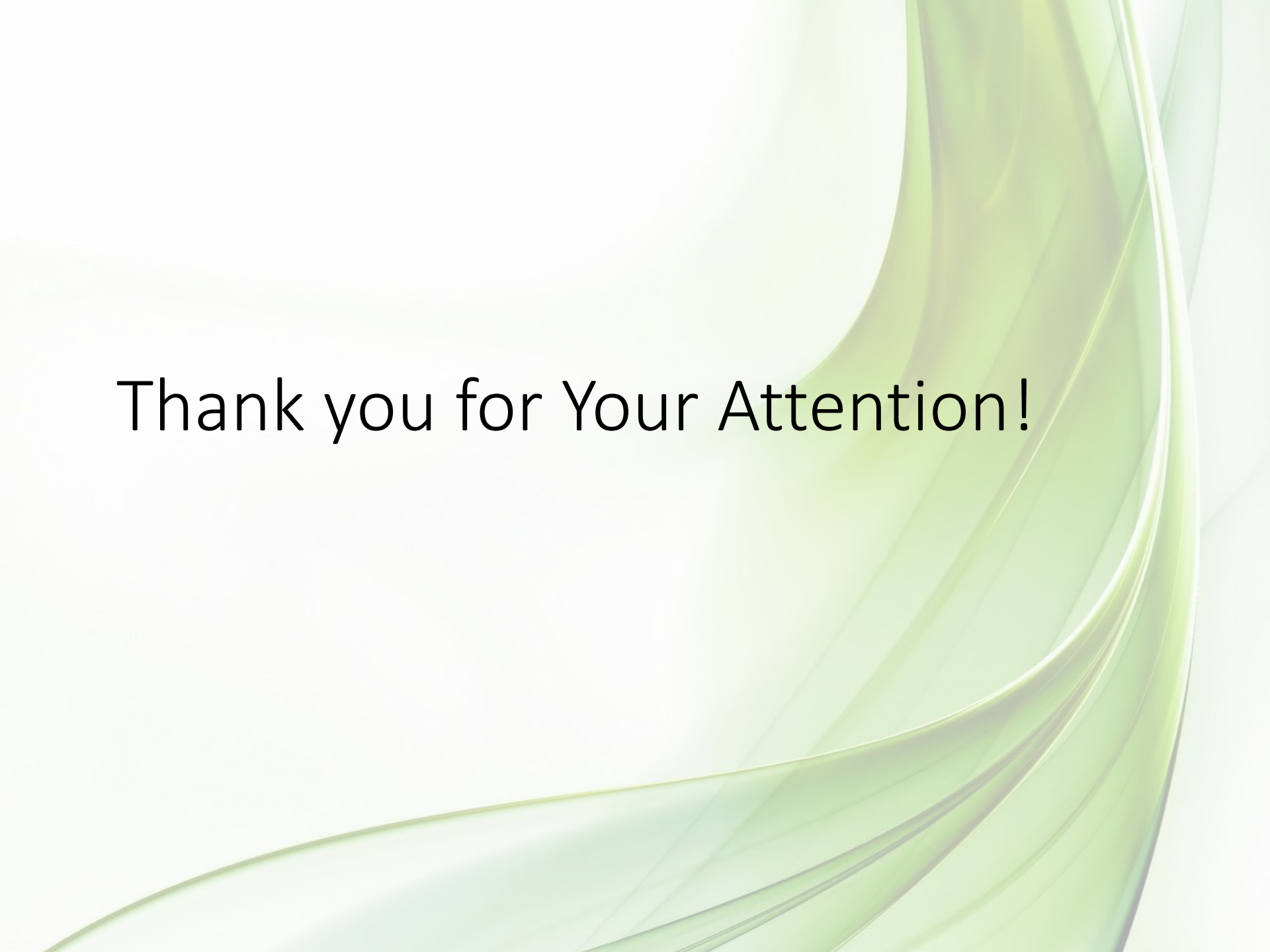


Tinbergen's Instrument-Targets Rule

- When we want to move from B to both internal and external equilibrium, the easiest way is to use of combination of two instruments
- General form of this conclusion: Jan Tinbergen
 - Several different formulations
 - For example: *there should be at least the same number of instruments as there are targets*

How about Greek situation during the Greek Crisis?



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Thank you for Your Attention!

References

- KOM: 17 + 18, 20
- Bahmani et al (2013): Empirical tests of the Marshall-Lerner condition: A literature review
- Frankel (2025): Resuscitating the Salter-Swan model of a small open economy
- Giudici & Lima (2026): Empirical analysis of the J-Curve and Marshall-Lerner condition: impacts of macroeconomic shocks on the Brazilian trade balance using a Bayesian SVAR Identified via heteroskedasticity